Project Title

Heart Disease Prediction System

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Abstract

Heart disease is one of the major health concerns worldwide and is responsible for many serious health complications. Early detection and proper risk assessment can help prevent complications and improve patient outcomes. Traditional methods of diagnosing heart disease involve medical examinations, laboratory tests, and expert evaluation, which may require considerable time and resources. With the development of Artificial Intelligence and Machine Learning, healthcare data can be analyzed more efficiently to identify patterns and support early disease risk assessment. This project presents a **Heart Disease Prediction System** developed using Python and Machine Learning techniques to predict the likelihood of heart disease based on important patient health and clinical parameters.

The dataset contains important attributes related to a patient's health, including age, gender, chest pain type, resting blood pressure, cholesterol level, fasting blood sugar, maximum heart rate, and other clinical factors. Before developing the prediction model, the data is cleaned and preprocessed to handle missing values and prepare it for machine learning algorithms. Exploratory Data Analysis and visualization techniques are used to understand the dataset and identify relationships between different features and heart disease. Python libraries such as **Pandas** and **NumPy** are used for data processing, while **Matplotlib** and **Seaborn** are used for visualization. The processed dataset is then used for training and testing the models.

Suitable machine learning classification algorithms are trained to predict whether a patient is likely to have heart disease or not. Their performance is evaluated using metrics such as accuracy, precision, recall, F1-score, and the confusion matrix. Based on the results, the best-performing model can be selected to make predictions using new patient information. The main objective is to demonstrate how Python, data analysis, and Machine Learning can support early risk assessment. However, the system is intended only as an educational and preliminary prediction tool and cannot replace professional medical diagnosis.